

Pearson Edexcel International GCSE ICT (4IT1)

Paper 1: Implications of Digital Technologies Mock Examination

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Total Marks: 150 Marks

Question 1: Data Protection, Individual Rights & Intellectual Property (Total: 40 marks)

Digital technologies have fundamentally changed how society shares and uses information, music, images, and films. While they provide powerful tools for development, they introduce significant societal challenges regarding legal data handling and asset protection.

(a) Identify the specific legal requirement mandated by Singapore's **Personal Data Protection Act** before an organization can collect, use, or disclose an individual's personal data. (1)

Enter legal requirement here...

(b) The United Kingdom's Data Protection Act restricts data use to strict data protection principles. Identify **four** distinct principles that organizations must adhere to when managing personal data records under this framework. (4)

- 1.
- 2.
- 3.

⁴(c) Under regional legislations like the UK Data Protection Act, citizens are granted specific statutory powers to preserve their data privacy.

State **three** individual legal rights available to a person whose private details are stored by an organization. (3)

- 1.
- 2.
- 3.

(d) User files are routinely stored across international border lines, causing complex jurisdiction conflicts.

(i) Describe the challenge that individuals face when their data is transferred to a country with weaker or non-existent data protection laws. (2)

Describe the jurisdiction security gap...

(ii) State the name of the specific international agreement framework designed to protect cross-border data transfers between the US and the European Union. (1)

Enter agreement framework name...

(e) Define the term **copyright** and state the immediate consequence a user faces if they download digital assets without permission. (3)

Definition and legal/operational consequence...

(f) Content creators have distinct models for sharing and licensing their assets.

(i) Contrast a paid licence with a free licence (such as Creative Commons or Open Source) in terms of distribution permissions. (2)

Contrast distribution rights here...

(ii) Explain how free software licences help reduce the digital divide and the information gap across global populations. (2)

Explain socioeconomic alignment impacts...

(g) In many countries, copyright exemptions (Fair Use) allow copying materials without the explicit permission of the owner under specific conditions.

Identify **four** legal exceptions or settings where copying copyrighted materials is lawful without direct permission. (4)

1.

2.

3.

⁴(h) Digital asset holders use technical barriers to maintain content boundaries.

Explain how Digital Rights Management (DRM) systems and geoblocking tools interact to check device authorizations and restrict files. (4)

Explain technical asset controls and verification pathways...

(i) State **three** types of digital information or assets that can be systematically protected by copyright laws. (3)

1.

2.

(j) State **three** actions or responsibilities organizations must carry out to ensure they act responsibly and protect the data they hold about individuals. (3)

1.

2.

3.

(k) Identify one country mentioned in the text that explicitly requires organizations to obtain informed, specific consent before using personal data. Place a cross [X] in the correct box. (1)

☐ A. United Kingdom

☐ B. Singapore

☐ C. Sweden

☐ D. Germany

(h) State one legal violation or condition under which an individual can claim financial compensation through courts under data protection laws. (1)

Enter statutory violation criteria...

(i) Explain why users should be able to give or withhold their explicit consent before a mobile application shares their real-time location. (2)

Explain consumer tracking consent parameters...

Question 2: Surveillance, Monitoring & Anti-Surveillance (Total: 30 marks)

Governments and commercial agencies systematically log movements and communications using diverse technological footprints.

(a) Match the specialized movement tracking mechanisms with their correct technical operational features: (5)

Mechanism	Sourced Technical Operational Feature Area
CCTV networks	
ANPR arrays	
Card Transactions	
Network Tracing	
GPS Technology	

(b) The widespread tracking of citizens remains highly controversial, involving balanced engineering and ethical considerations.

(i) Explain **three** practical benefits derived from monitoring individual or asset movements. (6)

Benefit 1:

Benefit 2:

Benefit 3:

(ii) Explain **three** clear structural drawbacks associated with establishing comprehensive movement-monitoring infrastructure. (6)

Drawback 1:

Drawback 2:

Drawback 3:

(c) To resist surveillance, modern fashion designers have developed countermeasure garments that function as camouflage.

Describe how anti-surveillance clothing and anti-surveillance eyewear work to disrupt or block tracking cameras. (4)

Anti-Surveillance Clothing:

Anti-Surveillance Eyewear:

(d) Digital interactions—including text messages, emails, and web histories—can be tracked under specific legal frameworks.

(i) Identify the software tool parents regularly deploy to track their children's online browsing activity. (1)

Enter structural family filter/tracking tool name...

(ii) State the name of the controversial UK legislation that legally permits security agencies to access digital communications, and identify the core human right it compromises. (3)

Legislation Name:

Human Right Compromised:

(e) State one direct benefit to an individual of having their communication paths monitored under lawful, protective guidelines. (1)

Enter single operational benefit...

(f) State one overarching drawback that monitoring both locations and communications imposes on a citizen's psychological well-being. (1)

Enter single psychological/behavioral impact...

Question 3: Sustainability & Environmental Impacts (Total: 25 marks)

The growing infrastructure of Information and Communication Technology (ICT) places an immense strain on global ecology.

(a) Define the terms **sustainability** and **mitigation** within an environmental context. (2)

Sustainability:

Mitigation:

(b) Complete the sustainability challenges matrix below by directly identifying the matching ecological problems and mitigation strategies. (12)

Cause of Issue	Sustainability / Ecological Problem	Sourced Mitigation Strategy
Power Requirements of Devices	Electricity generated using non-renewable sources depletes natural resources.	(i)
Cooling Systems in Data Centres	(ii)	(iii)
Use of Poisonous Substances	(iv)	(v)

(c) Improperly discarded electronic hardware fragments form a highly dangerous waste classification.

(i) State the technical term used for this specific type of electronic waste. (1)

Enter electronic waste vocabulary term...

(ii) Identify **three** toxic chemicals or substances contained inside digital components that can leak into soil and water structures. (3)

1. 2. 3.

(iii) List **three** severe human health conditions or organ damages caused when these toxic elements leak into surrounding community resources. (3)

1. 2. 3.

(d) Identify the strict international directive enacted by the European Union to manage the recycling of electronic components. (1)

Enter international recycling directive name...

(e) Explain why the large-scale use of online storage repositories and cloud applications can have a severe negative effect on the planet's environment. (3)

Explain backend data infrastructure footprint challenges...

Question 4: Ergonomics & Workplace Health and Safety (Total: 30 marks)

Prolonged or improper use of digital devices can lead to distinct occupational hazards affecting muscles, vision, and physical safety.

(a) Complete the following ergonomic risk management table by directly entering the missing causes or minimization methods from the textbook guidelines. (12)

Occupational Risk	Sourced Technical Cause	Sourced Minimization Method
Eye Dryness & Fatigue	Looking at screens for extended periods without blinking; exposure to short-wavelength blue light.	(a)
Repetitive Strain Injury (RSI)	(b)	Use ergonomic devices and cushioned wrist support pads.
Back & Neck Ache	Slouching or maintaining an incorrect sitting posture over extended periods.	(c)
Trip Hazards	(d)	Secure loose wires inside defensive plastic trunking channels.

(b) Modern office monitors emit short-wavelength light paths that interfere with biological human rhythms. Identify this specific light spectrum classification and state what biological loop it disrupts. (2)

Light Spectrum: Disrupted Loop:

(c) Review the diagnostic causes and minimization parameters for electrical safety inside a laboratory.

(i) Identify **two** distinct causes that trigger dangerous electric shocks at a workstation. (2)

- 1.
- 2.

(ii) State the name of the specialized circuit interrupting device utilized to prevent electric shock accidents. (1)

Enter circuit safety device term...

(iii) Identify **two** diagnostic causes that prompt dangerous hardware fire hazards inside a server closet. (2)

- 1.
- 2.

(iv) State **three** minimization safety setups a building safety warden must establish to suppress workstation fire threats. (3)

- 1.
- 2.
- 3.

(d) A sudden lack of environmental awareness due to severe mobile screen distractions can cause severe injury or death.

Identify **two** dangerous habits that trigger this loss of situational awareness and state how to minimize the risk. (3)

Habit 1: Habit 2:
Minimization Method:

(e) Analyze Figure 9.7 regarding optimal sitting postures at a computer terminal.

Identify **three** distinct structural posture settings an operator must align (e.g., thigh angle, keyboard alignment, or feet placement) to remain completely safe. (3)

- 1.
- 2.
- 3.

(f) Identify one activity that correctly reduces an individual's personal occupational risk when working in front of monitors. Place a cross [X] in the correct box. (1)

- ☐ A. Using a small monitor with low-level lighting
- ☐ B. Increasing the amount of short-wavelength blue light
- ☐ C. Integrating a Residual Current Device (RCD) into the circuit
- ☐ D. Maintaining a slouched posture to relax muscles

Question 5: Extended Response Essay — AI Automation & Ethical Risks (Total: 25 marks)

As artificial intelligence automation takes over daily human decisions—such as autopilot systems in airplanes or self-driving cars applying brakes, steering away from lane drift, and monitoring road speed limits—it introduces critical ethical dilemmas.

An automotive engineering firm is designing an autonomous transport network for a smart city. The technology board must evaluate how automated decision systems compute unexpected hazards compared to human vehicle operators.

Evaluate the implications of replacing human operators with algorithmic artificial intelligence systems. In your response, you must thoroughly evaluate:

- The Speed vs. Judgment Paradox (including processing speed parameters vs. rule-breaking flexibility).
- The Ethical Dilemma of moral accountability during an unavoidable multi-car accident or pedestrian collision.
- The Attention Lapse phenomenon and its effect on human reaction times when an operator is forced to take back manual control during a critical emergency.

Plan your answer carefully. Marks will be awarded for structured technical progression, correct usage of ICT vocabulary from the text, a balanced presentation of safety trade-offs, and an effective final conclusion.

Write essay answer within this container boundary layout blocks...

End of Question Bank